

# State Dependence in Union Membership

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## 1 Introduction

This study examines whether union membership persistence is driven by true state dependence— that is, whether past union membership directly increases the likelihood of current membership—or by unobserved individual characteristics. Previous studies highlights a difficulty in distinguishing between these effects (Vella and Verbeek, 1998) in regards to estimating the union premium for young men. We employ a CRE probit model and find that prior union membership increases the probability of current membership by approximately 8.4%. However we see that we might need further context for full dynamic completeness and consistency.

## 2 Data

We are use the following data set for time period 1980-84.

- *nr*- Person identifier
- *year* - Year (1981-1984) with lag
- *union* - 1 if in union, 0 otherwise
- *married* - 1 if married, 0 otherwise
- *educ* - Years of schooling
- *l1.union* - lagged variable, union membership in previous period
- *AvgMarried* - Avg married status for individual i.
- *union1* - initial union status.
- *yeardummies* - 1982,83,84

## 3 Econometric Theory

### Model

We consider a dynamic binary-choice model for union membership using a latent index formulation. The model is specified as:

$$y_{it}^* = \rho y_{i,t-1} + z_{it}\gamma + c_i + \epsilon_{it} \iff \text{union}_{it}^* = \rho \underbrace{\text{union}_{i,t-1}}_{\text{Lagged dependent variable}} + z_{it}\gamma + c_i + \epsilon_{it}$$

where  $\text{union}_{it} = 1[\text{union}_{it}^* > 0]$ . The key issue is that  $c_i$  (the unobserved individual effect) may be correlated with both the initial condition  $\text{union}_{i0}$  and the time-varying  $z_{it}$ .

## Correlated Random Effects

Because  $c_i$  might be correlated with the initial condition and the regressors, we employ a Correlated Random Effects (CRE) approach:

$$c_i = \psi + \xi_0 \text{union}_{i0} + \mathbf{z}_i \boldsymbol{\xi} + a_i, \quad a_i \sim N(0, \sigma_a^2)$$

where  $z_i$  represents the individual averages of the time-varying regressors. Substituting this into the latent model gives:

$$\text{union}_{it}^* = \rho \text{union}_{i,t-1} + \mathbf{z}_{it}\gamma + (\psi + \xi_0 \text{union}_{i0} + \mathbf{z}_i \boldsymbol{\xi} + a_i) + \epsilon_{it}$$

Rearranging:

$$\text{union}_{it}^* = \underbrace{(\psi + \xi_0 \text{union}_{i0} + \mathbf{z}_i \boldsymbol{\xi})}_{\text{CRE adjustment}} + \rho \text{union}_{i,t-1} + \mathbf{z}_{it}\gamma + a_i + \epsilon_{it}$$

The probability that  $y_{it} = 1$  is then:

$$P(\text{union}_{it} = 1 \mid \text{union}_{i,t-1}, \dots, \text{union}_{i0}, \mathbf{z}_i, c_i) = \Phi(\psi + \xi_0 \text{union}_{i0} + \mathbf{z}_i \boldsymbol{\xi} + \rho \cdot \text{union}_{i,t-1} + \mathbf{z}_{it}\gamma + a_i) \quad (1)$$

Since  $\epsilon_{it} \sim N(0, 1)$ , we assume a probit likelihood.

## Log-Likelihood Function

The model is estimated by Maximum Likelihood Estimation (MLE). We already know how the individual likelihood contribution look like and since  $a_i$  is unobserved, we integrate it out assuming a normal distribution. We do this by using the approximation of joint distribution  $\text{union}_i \mid z_i$  using Gauss-Hermite

$$L_i = \int_{-\infty}^{\infty} \prod_{t=1}^T \Phi\left(\psi + \xi_0 \text{union}_{i0} + \mathbf{z}_i \boldsymbol{\xi} + \rho \text{union}_{i,t-1} + \mathbf{z}_{it}\gamma + a_i\right)^{\text{union}_{it}} \left[1 - \Phi\left(\psi + \xi_0 \text{union}_{i0} + \mathbf{z}_i \boldsymbol{\xi} + \rho \text{union}_{i,t-1} + \mathbf{z}_{it}\gamma + a_i\right)\right]^{1-\text{union}_{it}} \frac{1}{\sqrt{2\pi}\sigma_a} e^{-\frac{a_i^2}{2\sigma_a^2}} da_i.$$

The log-likelihood is then maximized numerically over parameters  $\{\psi, \xi_0, \boldsymbol{\xi}, \rho, \gamma, \sigma_a\}$ .

## Average partial effects

An approach would be to use the average of our non-linear function instead of the non-linear function for the average (which is known as PEA) of each sample. This would

yield the average partial effect (APE), and this results in averaging the partial effects across the whole sample:

$$n^{-1} \sum_{i=1}^n g(\hat{\gamma}_0 + \mathbf{z}_i \hat{\gamma})$$

Since eq. (1) is our cumulative distribution function (CDF) in the probit model, we need to get the probability density function (PDF) as its derivative. We thus get the following:

$$g(z) = \frac{d\Phi(z)}{dz} = \phi(z) \quad \frac{\partial p(\mathbf{z})}{\partial z_j} = \phi(\mathbf{z}) \gamma_j \quad (2)$$

When writing the APE, we must also keep in mind that we eliminated  $a_i$  by integrating it out through Gaussian Quadrature. Thus, it's assumed that  $E[a_i] = 0$ . Inserting our information into the formula, we get the following expression for our APE:

$$APE_j = n^{-1} \sum_{i=1}^n \phi\left(\psi + \xi_{0\text{union}_{i0}} + \mathbf{z}_i \boldsymbol{\xi} + \rho_{\text{union}_{i,t-1}} + \mathbf{z}_{it} \boldsymbol{\gamma}\right) \gamma_j$$

## 4 Empirical analysis

### Model selection process

As we are aware of some individual specific time invariant effects  $c_i$  in our dynamic panel model, potentially causing spurious state dependence, meaning union membership appears to cause current membership, not because being in a union in the past directly influences current decisions, but rather because individuals differ persistently in unobserved ways that affect union membership decisions repeatedly over time. We have no choice to correct for  $c_i$  in order to assess the level of true state dependence. We assume strictly exogenous  $\mathbf{z}_{it}$ , assuming no significant feedback effects from union membership into our regressors years of education and marital status. We deem this a fair assumption. With this in mind we find a dynamic random effects model a suitable choice. However even with this approach we may encounter the initial conditions problem as  $c_i$  likely is correlated with  $\text{union}_{i0}$ . We extend the model further using the Chamberlain/Wooldridge approach, as defined in the theoretical section, assuming that the individual-specific effects  $c_i$  can be adequately approximated by the initial observed union membership  $\text{union}_0$  and individual averages of the strictly exogenous covariates. Other semiparametric approaches could be considered but does not serve our purpose of identifying the APE for our lagged variable. We therefore deem the CRE probit to be a good choice. To contextualise our results we also include the Pooled probit and RE-probit to see the effect on our APE for the lagged union variable as we do further controls for unobserved heterogeneity.

## Results

We conduct our 3 estimations, our CRE probit being our main model:

Table 1a (Pooled Probit) indicates strong state dependence with an average partial effect (APE) of lagged union membership at 0.397. However, ignoring individual-specific heterogeneity likely inflates this estimate.

Table 1b introduces a Random-Effects (RE) Probit model, controlling partially for unobserved heterogeneity. This reduces the lagged union APE slightly to 0.362, confirming the presence of persistent individual effects, as evidenced by a significant random-effects variance ( $\sigma_a = 0.494$ ).

Table 1c presents our preferred Correlated Random Effects (CRE) Probit model with Wooldridge's correction for initial conditions. Here, the estimated coefficient on lagged union membership drops significantly to 0.585, corresponding to an APE of 0.083.

This suggests that after properly accounting for unobserved individual heterogeneity and initial conditions, past union membership still increases the likelihood of current membership by about 8.3 percentage, though much less than suggested by simpler models. The initial union status coefficient is large (2.404) and strongly significant ( $APE = 0.340$ ), emphasizing the critical role initial conditions play in persistent union participation. Education shows a weak negative effect, while marital status is insignificant once averaged.

Table 1d provides a dynamic completeness test by including a lagged residual term. The strongly significant negative coefficient (1.852,  $t = 8.498$ ) clearly rejects dynamic completeness, implying residual dynamic structure not fully captured by the single-lag CRE specification.

## 5 Discussion of Assumptions

Correct Specification and Strict Exogeneity: Dynamic incompleteness suggests correlation between errors  $\varepsilon_{it}$  and regressors  $\mathbf{z}_{it}$ , potentially violating strict exogeneity due to omitted factors like economic conditions, biasing MLE estimates. Probit Errors: While we assume normally distributed, independent errors, systematic shocks (e.g., policy changes) may violate this. Random Effects Independence: Individual effects  $a_i$  are assumed independent, but unobserved worker preferences may still introduce bias despite our correction. Linearity of Individual Effects: Our linear specification may overlook interactions, though

significant initial union status suggests it captures key trends. No Additional Endogeneity: Rejection of dynamic completeness implies omitted variables (e.g., job changes) may cause inconsistency. Despite improved modeling, partial violations of key assumptions necessitate cautious interpretation of state dependence estimates.

## 6 Discussion and Conclusion

Our analysis suggests that there is a statistically and economically significant element of true state dependence in union membership. After controlling for unobserved heterogeneity and initial conditions, lagged union status increases the probability of current union membership by approximately eight percentage points. This suggests that being in a union before makes it more likely to stay in one, even after accounting for personal traits like a preference for unions or working in heavily unionized industries. However, since we fail dynamic completeness, part of this effect might still be due to other unmeasured factors changing over time.. Despite the improved specification, several limitations persist. First, the assumption that the unobserved individual effect enters additively and is normally distributed may omit more complex correlations between worker characteristics and union membership. For instance, a worker might experience a shift in job responsibilities or relocate to a region with different labor norms, introducing nonlinearities not captured by a linear correlated random effects formulation. Second, the rejection of dynamic completeness suggests that additional lag structures or time-varying unobserved components may affect membership decisions. Industries with shifting union influence or changing company policies can create patterns not captured by our model. Economic behavior also highlights potential deviations from our assumptions. If, for example, workers selectively switch to industries with stronger union traditions when they anticipate economic uncertainty, the randomness and normality assumptions on the error term may be violated. Overall the results underscore the importance of accounting for both unobserved heterogeneity and initial conditions when modeling union membership. We see that introducing further controls through going from a Pooled probit to FE-Probit to CRE-Probit, systematically lowers our APE for our lagged union membership variable indicating that these measure are somewhat effective, however our result of failing dynamic completeness may suggest this is still not enough. Further reserach could include more testing of strict exogeneity and experiments with different model specifications.

## Appendix: Tables and Figures

| Variable     | Estimate | Std. Error | t-value | APE    |
|--------------|----------|------------|---------|--------|
| Lagged Union | 1.857    | 0.088      | 20.992  | 0.397  |
| Married      | 0.088    | 0.071      | 1.245   | 0.019  |
| Education    | -0.010   | 0.019      | -0.543  | -0.002 |
| Year 1981    | -0.232   | Large      | -0.000  | -0.050 |
| Year 1982    | -0.199   | Large      | -0.000  | -0.043 |
| Year 1983    | -0.294   | Large      | -0.000  | -0.063 |
| Year 1984    | -0.248   | Large      | -0.000  | -0.053 |
| Constant     | -0.973   | Large      | -0.000  | -0.208 |

(a) Pooled probit estimates

| Variable             | Estimate          | Std. Error | t-value | APE    |
|----------------------|-------------------|------------|---------|--------|
| Lagged Union         | 0.585             | 0.093      | 6.304   | 0.083  |
| Initial Union Status | 2.404             | 0.178      | 13.476  | 0.340  |
| Married              | -0.069            | 0.141      | -0.492  | -0.010 |
| Avg. Married         | 0.284             | 0.175      | 1.621   | 0.040  |
| Education            | -0.052            | 0.028      | -1.825  | -0.007 |
| Year 1982            | 0.063             | 0.112      | 0.561   | 0.009  |
| Year 1983            | -0.020            | 0.097      | -0.202  | -0.003 |
| Year 1984            | 0.033             | 0.118      | 0.284   | 0.005  |
| Constant             | -1.417            | 0.321      | -4.414  | -0.201 |
| $\sigma_a$           | 0.842             | 0.056      | 15.169  | 0.119  |
| Observations         | 2180 (545 groups) |            |         |        |
| Log-Likelihood       | -693.470          |            |         |        |

(c) RE probit with Wooldridge/Chamberlain correction

| Variable     | Estimate | Std. Error | t-value | APE    |
|--------------|----------|------------|---------|--------|
| Lagged Union | 1.685    | 0.227      | 7.414   | 0.362  |
| Married      | 0.093    | 0.070      | 1.330   | 0.020  |
| Education    | -0.013   | 0.020      | -0.660  | -0.003 |
| Year 1981    | -0.213   | 0.073      | -2.914  | -0.046 |
| Year 1982    | -0.200   | 0.058      | -3.469  | -0.043 |
| Year 1983    | -0.309   | 0.067      | -4.605  | -0.066 |
| Year 1984    | -0.264   | 0.072      | -3.678  | -0.057 |
| Constant     | -0.986   | 0.186      | -5.305  | -0.212 |
| $\sigma_a$   | 0.494    | 0.262      | 1.881   | 0.106  |

(b) RE probit estimates (uncorrected)

| Variable        | Estimate          | Std. Error | t-value | APE    |
|-----------------|-------------------|------------|---------|--------|
| Lagged Union    | 3.349             | 0.202      | 16.598  | 0.638  |
| Married         | -0.053            | 0.177      | -0.300  | -0.010 |
| Avg. Married    | 0.128             | 0.198      | 0.648   | 0.024  |
| Education       | -0.030            | 0.018      | -1.660  | -0.006 |
| Year 1982       | -0.300            | 0.082      | -3.641  | -0.057 |
| Year 1983       | -0.398            | 0.081      | -4.901  | -0.076 |
| Year 1984       | -0.331            | 0.075      | -4.396  | -0.063 |
| Lagged Residual | -1.852            | 0.218      | -8.498  | -0.353 |
| Constant        | -1.029            | 0.170      | -6.062  | -0.196 |
| Observations    | 1635 (545 groups) |            |         |        |
| Log-Likelihood  | -569.265          |            |         |        |

(d) Dynamic completeness test

Table 1: Overview of different estimators